

Aditya Kulkarni

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Raleigh, NC (open to relocating to San Francisco)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience designing and deploying AI agent systems. At Isomer AI, built Temporal.io workflow orchestration and multi-model AI agent tooling across Claude, GPT-4o, and Gemini using MCP -- 394 commits, 7 production deployments on a live claims automation product. Independently built AlgoPulse, a fault-tolerant agentic platform processing 2,000 async inference requests/second, and Glance AI, a compliance-ready RAG pipeline with per-tenant data isolation for fintech and healthcare. Case studies: kulkarniaditya.com

CORE COMPETENCIES

AI Agent Development

Agentic Workflow Orchestration

Production LLM Systems

RAG Pipeline Design

Temporal.io & Kafka

TypeScript / Node.js

Multi-model LLM Routing

Eval Frameworks & Observability

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built **Temporal.io workflow orchestration** and multi-model AI agent tooling across 3 LLM providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models using MCP -- core of a production AI agent system for claims automation across 7 production deployments
- Shipped 10+ backend features including role-based access control, real-time event pipelines (Pusher), and document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) -- **394 commits** to a live TypeScript / Node.js / PostgreSQL / Redis codebase
- Integrated Auth0, Docker, Playwright, and Sentry for authentication, containerization, end-to-end testing, and production observability across the agent platform

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform with JBrowse 2, serving 500GB of strawberry and grape datasets to 100+ researchers worldwide, deployed on Google Cloud Run
- Applied hyperparameter tuning and time-series ML analysis, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML pipelines; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated Grafana monitoring dashboards; supported model validation and red-teaming to improve production reliability

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a support ticket analysis system; built distributed data pipelines with Hadoop + Spark and Kafka streaming -- reduced ticket-processing latency by 20%

PROJECTS

AlgoPulse -- Agentic AI Platform Independent, 2025

Architected a fault-tolerant, event-driven AI orchestration platform. Identified API-polling bottleneck under load; pivoted to Supabase Edge Functions streaming AI reasoning loops to clients. Temporal.io manages durable workflow state; Kafka handles worker coordination and message brokering. Scaled to **2,000 async inference requests/second** without state loss -- zero message loss under load.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Privacy-First RAG Pipeline Independent, 2025

Built a compliance-ready RAG system for fintech and healthcare environments with per-tenant data isolation at the vector retrieval layer. Prevents cross-tenant context leakage and LLM hallucinations; deterministic output architecture ensures factual correctness in regulated data environments.

Python, LLMs, Vector Databases, RAG

CryptoKnight -- Trustless AI Trading Agent HackNC State, 2025

Solved the "Delegation Dilemma" for autonomous AI agents: built a trustless architecture using ERC-4337 Account Abstraction so an AI agent trades Polymarket prediction markets without holding private keys. Session Keys enforce on-chain policies -- trade limits, allowance cap, expiration, instant revocation -- agent acts autonomously within cryptographically enforced constraints.

ERC-4337, Alchemy Account Kit, Python agent, Next.js 16, viem, wagmi

PerCent -- GenAI Finance App HackNC24 Top 10 / 100+ teams

Built a RAG-based conversational AI chatbot using Qwen-14B-Chat for financial domain reasoning; reduced manual research time by 30%. Shipped complete MVP in 48 hours -- goal tracking, expense logging, AI chatbot -- ranked Top 10 of 100+ teams.

Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

EDUCATION

Master of Science, Computer Science North Carolina State University

Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech, Computer Science Mumbai University

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Machine Learning, Big Data Infrastructure, Blockchain, Algorithms

SKILLS

Languages: TypeScript, Python, JavaScript, Java, SQL, C

AI/ML: LLM APIs (Claude, GPT-4o, Gemini), MCP, LangChain, RAG, Hugging Face, TensorFlow, PyTorch, Prompt Engineering

Agent Infra: Temporal.io, Kafka, Supabase Edge Functions, Distributed Systems

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry

Frameworks: Node.js, React, Express, Prisma, PostgreSQL, Redis, Auth0